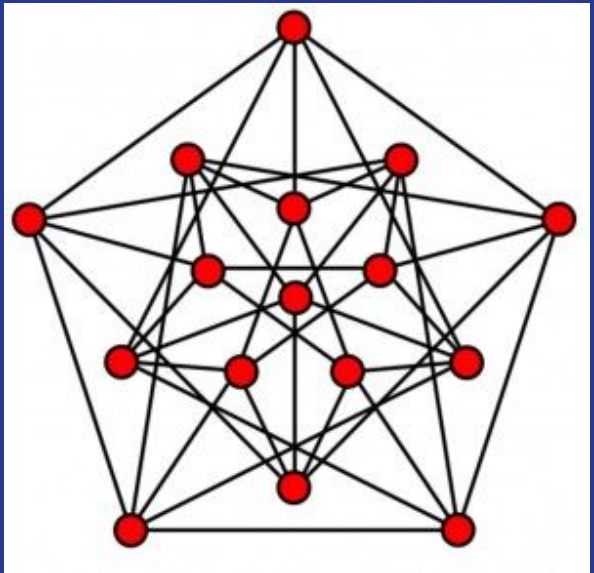
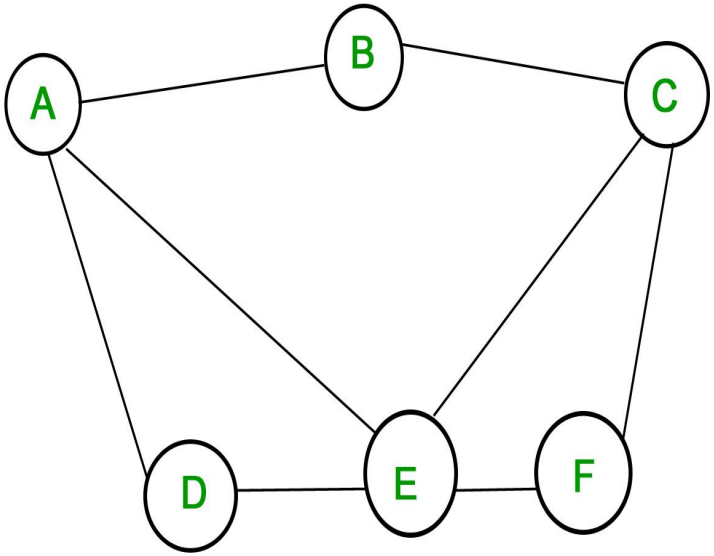
Turán Number for a *k*-Chorded Cycle

By: Vaibhav Mishra, Gautham Anne
Mentor: Dr. Brad Elliott



What is a Graph?

- Graph
- Vertex
- Edge
- Cycle
- Chorded Cycle
- Doubly Chorded Cycle
- Triply Chorded Cycle
- Degree/Minimum Degree
- Leaf
- Subgraph



Problem

- A Turán number is the maximum number of edges a graph can have without containing a certain subgraph.

- The subgraphs that we are working with are chorded cycles (one-chorded, doubly-chorded, triply-chorded, etc.).

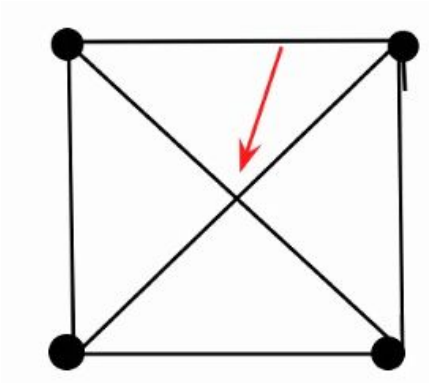
- What is the Turán number for a *k*-chorded cycle?

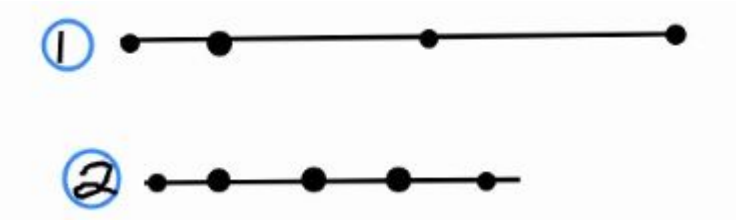
Solution

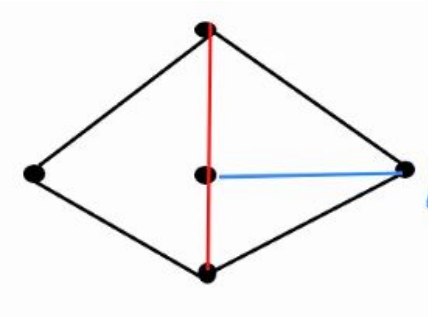
- Try out smaller values of k (k is the number of chords)
- Try it for graphs with small # of vertices
- Find a pattern and try to generalize and prove for all values of k.

Common Misconceptions

- Vertices of intersecting edges
- Longest Path in a Graph
- No chord when vertex-> vertex-> vertex





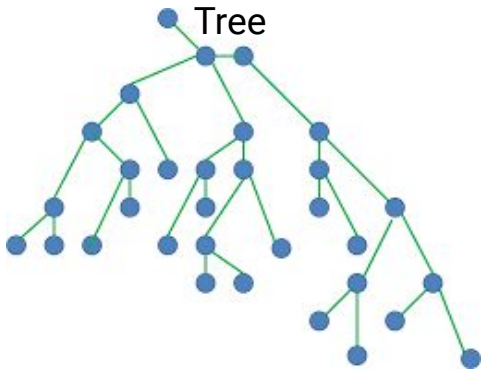


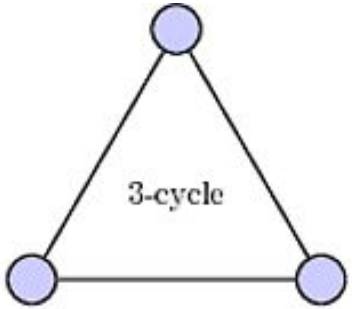
Longest Path Theorem

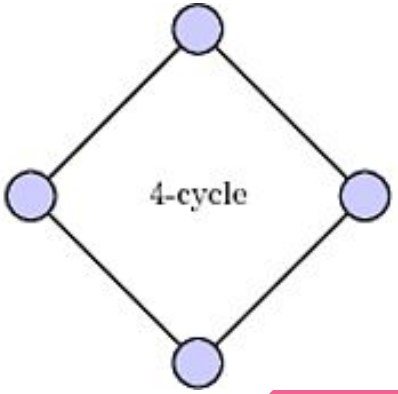
In the longest path of n vertices where the minimum degree of the path is two, all vertices must have at least two edges that come out from it.

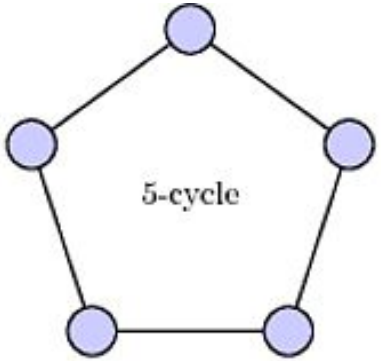
Turán Number for A Cycle

- Lemma 1: A graph with minimum degree two must have a cycle.
- Theorem 1: A graph with the same number of vertices and edges contains a cycle.
- Conclusion: Turán Number for a Cycle is n-1.



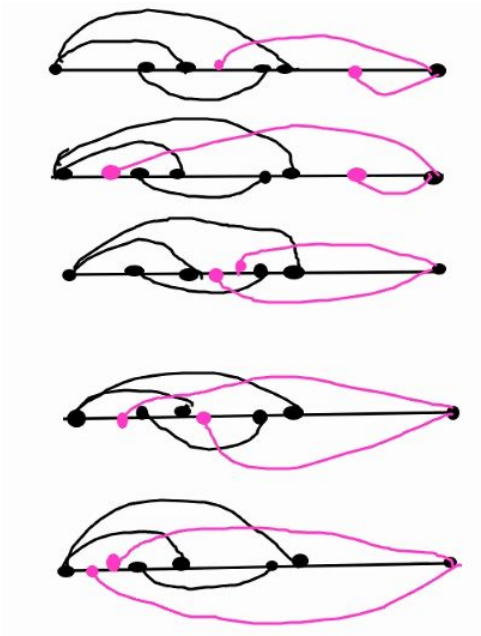






Common Methodologies

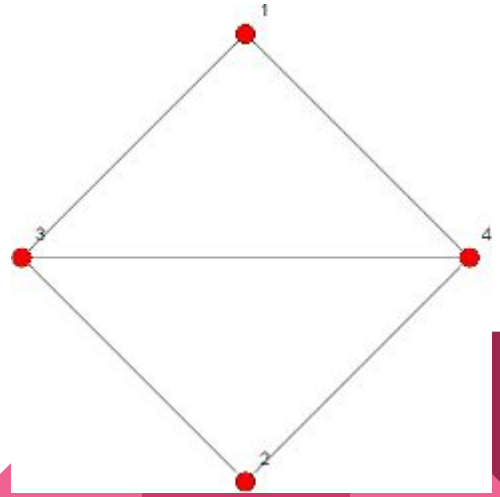
- Casework
- Proof by contradiction (Assume outcome X and then prove X wrong)
- Graph Decomposition
- Longest Path Method
- Extremal Graphs



Turán Number for A Chorded Cycle

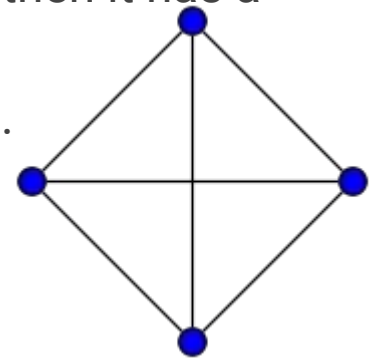
- Definition: Complete Bipartite Graph
- Lemma 2: If a graph has minimum degree three, then it has a chorded cycle.
- Theorem 2: If graph has n vertices, and at least 2n-3 edges then it has a chorded cycle.
- Conclusion: Turán number for a chorded cycle is 2n-4.

Solved by: Graph Decomposition



Turán Number for a Doubly Chorded Cycle

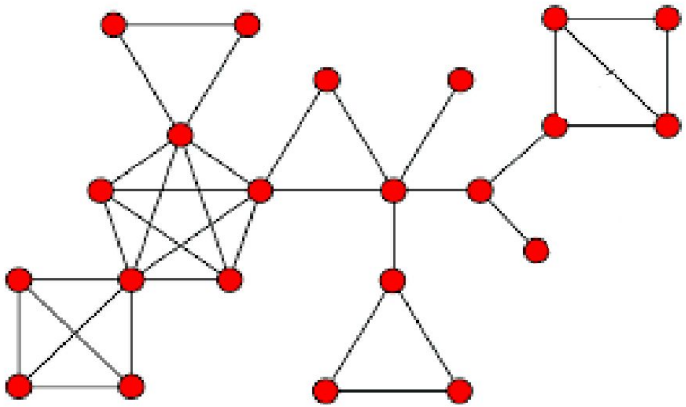
- Lemma 3: If a graph has minimum degree 3, then it has a doubly chorded cycle.
- Theorem 3: If a graph has n vertices and at least 2n-2 edges, then it has a doubly chorded cycle.
- Conclusion: Turán Number for a doubly chorded cycle is 2n-3.



Solved by: Casework and the Longest Path Theorem

Block

- Definition of a Block: A maximal subgraph where every pair of vertices lie together on some cycle.
- An individual edge is also considered a block even though it has no cycle.



Turán Number for a Triply Chorded Cycle

- Lemma 4: If a graph has n vertices and at least 2n-1 edges, then it must have a block with k vertices and 2k-1 edges.
- Theorem 4: If a graph has n vertices and at least **2n-1** edges, then it must have a **triply** chorded cycle.
- Conclusion: The Turán number for a triply chorded cycle is 2n - 2.

Partially-Solved by: Contradiction and Casework

Conjecture (Turán Number for a *k*-chorded cycle)

Number of Chords	Turán Number
1-chorded cycle	2n-4
2-chorded cycle	2n-3
3-chorded cycle	2n-2
4-chorded cycle	????

Future Work and Study

- Find the Turán number for a 4-chorded cycle, 5-chorded cycle, 6-chorded cycle…
- General formula for a *k*-chorded cycle
- How subgraphs can be found in other graphs of n vertices, e edges, B blocks, etc.

Acknowledgements

We would like to thank our mentor Dr. Elliott for teaching and guiding us through the research process. We would also like to thank the Math Circle program for allowing us to venture into mathematics with their three-week research program.